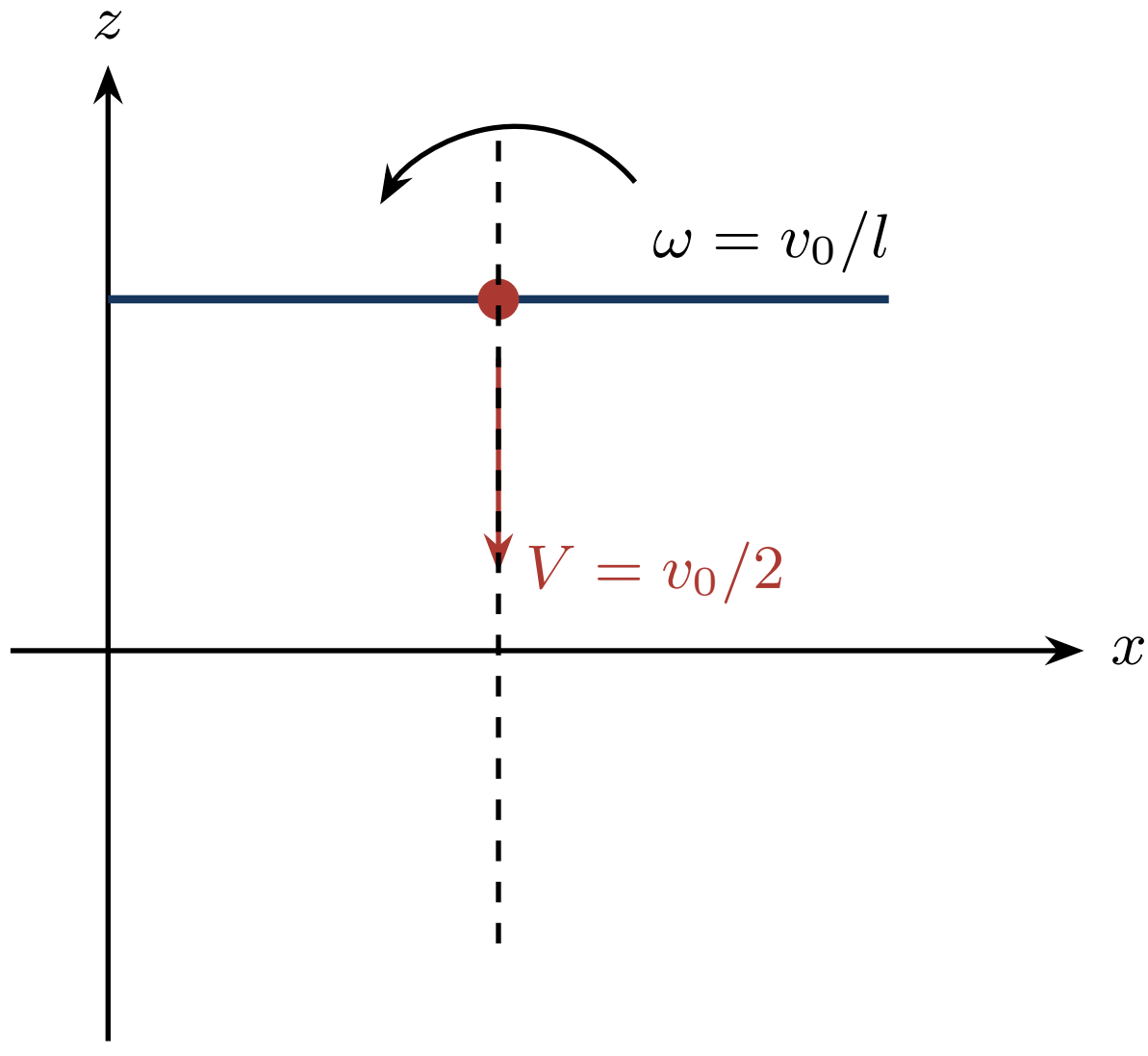

局部下落：质心匀速向下，笔绕质心逆时针转动